Практика 5: Детальная настройка привилегий для ранее созданных ролей

1.1

```sql
-- 2. Текущие права на таблицы
SELECT
    grantee AS "Роль",
    table_schema AS "Схема",
    table_name AS "Таблица",
    privilege_type AS "Привилегия",
    is_grantable AS "Может передавать"
FROM information_schema.role_table_grants
WHERE grantee IN (
    'app_user', 'app_manager', 'app_admin'
)
ORDER BY grantee, table_name, privilege_type;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	Роль	Схема	Таблица	Привилегия	Может передавать
1	app_admin	app	access_logs	DELETE	NO
2	app_admin	app	access_logs	INSERT	NO
3	app_admin	app	access_logs	REFERENCES	NO
4	app_admin	app	access_logs	SELECT	NO
5	app_admin	app	access_logs	TRIGGER	NO
6	app_admin	app	access_logs	TRUNCATE	NO
7	app_admin	app	access_logs	UPDATE	NO
8	app_admin	app	comments	DELETE	NO
9	app_admin	app	comments	INSERT	NO
10	app_admin	app	comments	REFERENCES	NO
11	app_admin	app	comments	SELECT	NO
12	app_admin	app	comments	TRIGGER	NO
13	app_admin	app	comments	TRUNCATE	NO
14	app_admin	app	comments	UPDATE	NO
15	app_admin	app	departments	DELETE	NO
16	app_admin	app	departments	INSERT	NO
17	app_admin	app	departments	REFERENCES	NO
18	app_admin	app	departments	SELECT	NO
19	app_admin	app	departments	TRIGGER	NO
20	app_admin	app	departments	TRUNCATE	NO
21	app_admin	app	departments	UPDATE	NO

	Роль	Схема	Таблица	Привилегия	Может передавать
22	app_admin	app	positions	DELETE	NO
23	app_admin	app	positions	INSERT	NO
24	app_admin	app	positions	REFERENCES	NO
25	app_admin	app	positions	SELECT	NO
26	app_admin	app	positions	TRIGGER	NO
27	app_admin	app	positions	TRUNCATE	NO
28	app_admin	app	positions	UPDATE	NO
29	app_admin	app	projects	DELETE	NO
30	app_admin	app	projects	INSERT	NO
31	app_admin	app	projects	REFERENCES	NO
32	app_admin	app	projects	SELECT	NO
33	app_admin	app	projects	TRIGGER	NO
34	app_admin	app	projects	TRUNCATE	NO
35	app_admin	app	projects	UPDATE	NO
36	app_admin	app	task_history	DELETE	NO
37	app_admin	app	task_history	INSERT	NO
38	app_admin	app	task_history	REFERENCES	NO
39	app_admin	app	task_history	SELECT	NO
40	app_admin	app	task_history	TRIGGER	NO
41	app_admin	app	task_history	TRUNCATE	NO
42	app_admin	app	task_history	UPDATE	NO
43	app_admin	app	tasks	DELETE	NO
44	app_admin	app	tasks	INSERT	NO
45	app_admin	app	tasks	REFERENCES	NO
46	app_admin	app	tasks	SELECT	NO
47	app_admin	app	tasks	TRIGGER	NO
48	app_admin	app	tasks	TRUNCATE	NO
49	app_admin	app	tasks	UPDATE	NO
50	app_admin	app	users	DELETE	NO
51	app_admin	app	users	INSERT	NO
52	app_admin	app	users	REFERENCES	NO
53	app_admin	app	users	SELECT	NO

54	app_admin	app	users	TRIGGER	NO
55	app_admin	app	users	TRUNCATE	NO
56	app_admin	app	users	UPDATE	NO
57	app_manager	app	access_logs	SELECT	NO
58	app_manager	app	comments	DELETE	NO
59	app_manager	app	comments	INSERT	NO
60	app_manager	app	comments	SELECT	NO
61	app_manager	app	comments	UPDATE	NO
62	app_manager	app	departments	SELECT	NO
63	app_manager	app	positions	SELECT	NO
64	app_manager	app	projects	INSERT	NO
65	app_manager	app	projects	SELECT	NO
66	app_manager	app	projects	UPDATE	NO
67	app_manager	app	task_history	INSERT	NO
68	app_manager	app	task_history	SELECT	NO
69	app_manager	app	tasks	INSERT	NO
70	app_manager	app	tasks	SELECT	NO
71	app_manager	app	tasks	UPDATE	NO
72	app_manager	app	users	SELECT	NO
73	app_user	app	users	SELECT	NO

```
2   -- 3. Права на схемы
3   SELECT
4       nspname AS "Схема",
5       nspacl AS "Права"
6   FROM pg_namespace
7   WHERE nspname IN ('public', 'app');
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	Схема	Права
1	public	{pg_database_owner=UC/pg_database_owner,=U/pg_data…
2	app	{postgres=UC/postgres,app_connect=U/postgres,app_a…

```sql
-- 4. Права на базу данных
SELECT
    datname AS "База данных",
    datacl AS "Права"
FROM pg_database
WHERE datname = 'corporate_tasks';
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	База данных	Права
1	corporate_tasks	{=Tc/postgres,postgres=CTc/postgres,app_connect=c/…

```sql
-- 5. Последовательности
SELECT
    schemaname,
    sequencename,
    sequenceowner
FROM pg_sequences
WHERE schemaname = 'app';
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	schemaname	sequencename	sequenceowner
1	app	departments_department_id_seq	postgres
2	app	positions_position_id_seq	postgres
3	app	users_user_id_seq	postgres
4	app	projects_project_id_seq	postgres
5	app	tasks_task_id_seq	postgres
6	app	comments_comment_id_seq	postgres
7	app	task_history_history_id_seq	postgres
8	app	access_logs_log_id_seq	postgres

В таблице на текущие права есть несоответствие. По сравнению с таблицей из практики 4

У **app_manager** есть доступ к просмотру журнала доступа

У **app_user** есть доступ к просмотру всей таблице пользователей

2.1

```
3   -- Отзыв всех прав на таблицы
4
5   REVOKE ALL PRIVILEGES ON ALL TABLES IN SCHEMA app
6   FROM app_user, app_manager, app_admin;
7
```

Problems Output Debug Console Terminal Ports PostgreSQL Quer

Messages

```
8:15:45 PM     Started executing query at Line 2
               REVOKE
               Total execution time: 00:00:00.006
```

```
2   -- Отзыв всех прав на последовательности
3
4   REVOKE ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA app
5   FROM app_user, app_manager, app_admin;
6
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Resul

Messages

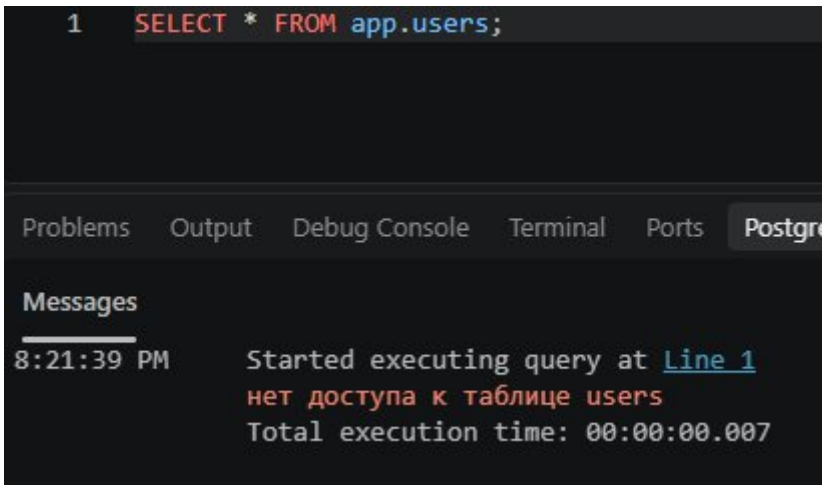
```
:16:31 PM     Started executing query at Line 2
              REVOKE
              Total execution time: 00:00:00.001
```

```
2   -- Проверка оставшихся прямых прав
3
4   SELECT
5       grantee,
6       table_name,
7       privilege_type
8   FROM information_schema.role_table_grants
9   WHERE grantee IN (
10      'app_user',
11      'app_manager',
12      'app_admin'
13  );
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Resul

Results Messages

grantee	table_name	privilege…



Проверка через app_user

2.2

```sql
-- 1. Базовое подключение
-- app_connect уже существует
GRANT CONNECT ON DATABASE corporate_tasks TO app_connect;
GRANT USAGE ON SCHEMA app TO app_connect;

-- 2. Чтение справочников
-- app_read_reference уже существует
GRANT SELECT ON TABLE app.departments TO app_read_reference;
GRANT SELECT ON TABLE app.positions TO app_read_reference;

-- 3. Чтение основных таблиц
CREATE ROLE app_read_main NOLOGIN;

GRANT SELECT ON TABLE app.users TO app_read_main;
GRANT SELECT ON TABLE app.projects TO app_read_main;
GRANT SELECT ON TABLE app.tasks TO app_read_main;

-- 4. Полные права на комментарии
CREATE ROLE app_comments_full NOLOGIN;

GRANT SELECT, INSERT, UPDATE, DELETE
ON TABLE app.comments TO app_comments_full;

GRANT USAGE ON SEQUENCE app.comments_comment_id_seq
TO app_comments_full;

-- 5. Управление задачами
CREATE ROLE app_task_write NOLOGIN;

GRANT SELECT, INSERT, UPDATE
ON TABLE app.tasks TO app_task_write;

GRANT USAGE ON SEQUENCE app.tasks_task_id_seq
TO app_task_write;

GRANT SELECT ON TABLE app.projects TO app_task_write;

-- 6. Управление проектами
CREATE ROLE app_project_write NOLOGIN;

GRANT SELECT, INSERT, UPDATE
ON TABLE app.projects TO app_project_write;

GRANT USAGE ON SEQUENCE app.projects_project_id_seq
TO app_project_write;
```

```sql
-- 7. История изменений
CREATE ROLE app_history_full NOLOGIN;

GRANT SELECT, INSERT
ON TABLE app.task_history TO app_history_full;

GRANT USAGE ON SEQUENCE app.task_history_history_id_seq
TO app_history_full;

-- 8. Запись в журнал доступа
CREATE ROLE app_access_log_write NOLOGIN;

GRANT INSERT ON TABLE app.access_logs TO app_access_log_write;

GRANT USAGE ON SEQUENCE app.access_logs_log_id_seq
TO app_access_log_write;

-- 9. Полный доступ для администратора
CREATE ROLE app_full_access NOLOGIN;

GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA app
TO app_full_access;

GRANT ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA app
TO app_full_access;

GRANT CREATE ON SCHEMA app TO app_full_access;
```

```sql
-- Добавление комментариев к ролям

COMMENT ON ROLE app_connect
IS 'Базовое подключение к БД';

COMMENT ON ROLE app_read_reference
IS 'Чтение справочников';

COMMENT ON ROLE app_read_main
IS 'Чтение основных таблиц';

COMMENT ON ROLE app_comments_full
IS 'Полный доступ к комментариям';

COMMENT ON ROLE app_task_write
IS 'Чтение и запись задач';

COMMENT ON ROLE app_project_write
IS 'Чтение и запись проектов';

COMMENT ON ROLE app_history_full
IS 'Чтение и запись истории изменений';

COMMENT ON ROLE app_access_log_write
IS 'Запись в журнал доступа';

COMMENT ON ROLE app_full_access
IS 'Полный доступ к объектам приложения';
```

```sql
-- Просмотр ролей и комментариев

SELECT
    rolname,
    rolcanlogin,
    pg_catalog.shobj_description(
        oid, 'pg_authid'
    ) AS description
FROM pg_roles
WHERE rolname LIKE 'app_%'
ORDER BY rolname;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	rolname	rolcanlogin	description
1	app_access_log_write	False	Запись в журнал доступа
2	app_admin	True	*NULL*
3	app_audit_read	False	*NULL*
4	app_comments_full	False	Полный доступ к комментариям
5	app_connect	False	Базовое подключение к БД
6	app_employee	False	*NULL*
7	app_full_access	False	Полный доступ к объектам приложения
8	app_guest	False	*NULL*
9	app_history_full	False	Чтение и запись истории изменений
10	app_history_read	False	*NULL*
11	app_internal_comments	False	*NULL*
12	app_manager	True	*NULL*
13	app_project_write	False	Чтение и запись проектов
14	app_read_all	False	*NULL*
15	app_read_main	False	Чтение основных таблиц
16	app_read_reference	False	Чтение справочников
17	app_superuser	False	*NULL*
18	app_task_worker	False	*NULL*
19	app_task_write	False	Чтение и запись задач
20	app_user	True	*NULL*

3.1

```sql
-- Задание 3.1. Настройка роли app_user

-- Базовое подключение
GRANT app_connect TO app_user;

-- Чтение справочников
GRANT app_read_reference TO app_user;

-- Чтение основных таблиц
GRANT app_read_main TO app_user;

-- Чтение и изменение задач
GRANT app_task_write TO app_user;

-- Работа с комментариями
GRANT app_comments_full TO app_user;

-- Чтение и запись истории
GRANT app_history_full TO app_user;

-- Запись в журнал доступа
GRANT app_access_log_write TO app_user;
```

```sql
-- Проверка через information_schema

SELECT table_name, privilege_type
FROM information_schema.role_table_grants
WHERE grantee = 'app_user'
ORDER BY table_name, privilege_type;
```

Problems Output Debug Console Terminal Ports Post

Results Messages

table_name	privilege…

```sql
SELECT * FROM app.departments;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... |

	department_id	name	description	created_at
1	1	IT Department	Information Technology	2026-11-03 20:14:12.844471
2	2	Marketing	Marketing and Communications	2026-11-03 20:14:12.844471
3	3	Sales	Sales Department	2026-11-03 20:14:12.844471

3.2

```sql
-- Задание 3.2. Настройка app_manager

-- Менеджер наследует права обычного пользователя
GRANT app_user TO app_manager;

-- Дополнительные права на управление проектами
GRANT app_project_write TO app_manager;

-- Права на столбец внутренних комментариев
GRANT SELECT (is_internal), UPDATE (is_internal)
ON TABLE app.comments TO app_manager;
```

```sql
SELECT table_name, privilege_type
FROM information_schema.role_table_gran
WHERE grantee = 'app_manager'
ORDER BY table_name, privilege_type;
```

Problems Output Debug Console Terminal Ports

Results Messages

table_name	privilege…

```sql
-- Проверяем чтение справочников
SELECT * FROM app.departments;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... |

	department_id	name	description	created_at
1	1	IT Department	Information Technology	2026-11-03 20:14:12.844471
2	2	Marketing	Marketing and Communications	2026-11-03 20:14:12.844471
3	3	Sales	Sales Department	2026-11-03 20:14:12.844471

3.3

```
-- Задание 3.3. Настройка app_admin

-- Администратор наследует права менеджера
GRANT app_manager TO app_admin;

-- Полный доступ к объектам приложения
GRANT app_full_access TO app_admin;

-- Право создавать объекты в схеме app
GRANT CREATE ON SCHEMA app TO app_admin;
```

```
-- Проверка прав администратора

SELECT table_name, privilege_type
FROM information_schema.role_table_grants
WHERE grantee = 'app_admin'
ORDER BY table_name, privilege_type;
```

Problems Output Debug Console Terminal Ports PostgreSC

Results Messages

table_name	privilege…

```
-- Проверка чтения журнала доступа
SELECT * FROM app.access_logs;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... | corporate_ta... | schema: session default

log_id	user_id	action	resource_…	resource_…	ip_address	logged_at

4.1

```sql
-- 1. Чтение новых таблиц
ALTER DEFAULT PRIVILEGES FOR ROLE app_admin
IN SCHEMA app
GRANT SELECT ON TABLES TO app_read_reference;

-- 2. Работа с новыми таблицами
ALTER DEFAULT PRIVILEGES FOR ROLE app_admin
IN SCHEMA app
GRANT SELECT, INSERT, UPDATE, DELETE
ON TABLES TO app_task_write;

-- 3. Полные права для администратора
ALTER DEFAULT PRIVILEGES FOR ROLE app_admin
IN SCHEMA app
GRANT ALL ON TABLES TO app_full_access;

-- 4. Использование новых последовательностей
ALTER DEFAULT PRIVILEGES FOR ROLE app_admin
IN SCHEMA app
GRANT USAGE ON SEQUENCES TO app_task_write;

-- 5. Полные права на последовательности
ALTER DEFAULT PRIVILEGES FOR ROLE app_admin
IN SCHEMA app
GRANT ALL ON SEQUENCES TO app_full_access;
```

```sql
-- Проверка DEFAULT PRIVILEGES

SELECT
    defaclrole::regrole AS "Роль-создатель",
    defaclnamespace::regnamespace AS "Схема",
    defaclobjtype AS "Тип объекта",
    defaclacl AS "Права по умолчанию"
FROM pg_default_acl
WHERE defaclnamespace::regnamespace::text = 'app';
```

oblems Output Debug Console Terminal Ports **PostgreSQL Query Results**

esults Messages localho... | corporate_ta... | sche

Роль-создатель	Схема	Тип объекта	Права по умолчанию
app_admin	app	r	{app_read_reference=r/app_admin,app_task_write=arw…
app_admin	app	S	{app_task_write=U/app_admin,app_full_access=rwU/ap…

```sql
-- Создание таблицы от имени app_admin

SET ROLE app_admin;

CREATE TABLE app.test_default_privileges (
    id SERIAL PRIMARY KEY,
    name TEXT,
    description TEXT,
    created_at TIMESTAMP DEFAULT NOW()
);

RESET ROLE;
```

```sql
-- Права на новую таблицу

SELECT
    grantee,
    privilege_type
FROM information_schema.role_table_grants
WHERE table_name = 'test_default_privileges'
ORDER BY grantee, privilege_type;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Result

Results Messages

	grantee	privilege_type
1	app_admin	DELETE
2	app_admin	INSERT
3	app_admin	REFERENCES
4	app_admin	SELECT
5	app_admin	TRIGGER
6	app_admin	TRUNCATE
7	app_admin	UPDATE
8	app_full_access	DELETE
9	app_full_access	INSERT
10	app_full_access	REFERENCES
11	app_full_access	SELECT
12	app_full_access	TRIGGER
13	app_full_access	TRUNCATE
14	app_full_access	UPDATE
15	app_read_reference	SELECT
16	app_task_write	DELETE
17	app_task_write	INSERT
18	app_task_write	SELECT
19	app_task_write	UPDATE

```sql
-- Проверка добавления записи

INSERT INTO app.test_default_privileges (name)
VALUES ('Test default privileges');

```

Добавление записи за пользователя

```sql
DROP TABLE app.test_default_privileges;
```

Удаление таблицы

5.1

```sql
-- Тест 1: Чтение справочников
SELECT * FROM app.departments;
SELECT * FROM app.positions;

-- Тест 2: Чтение пользователей
SELECT user_id, username, email, full_name
FROM app.users;

-- Тест 3: Чтение проектов
SELECT * FROM app.projects;

-- Тест 4: Чтение задач
SELECT * FROM app.tasks;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... | corporate_

	department_id	name	description	created_at
1	1	IT Department	Information Technology	2026-11-03 20:14:12.844471
2	2	Marketing	Marketing and Communications	2026-11-03 20:14:12.844471
3	3	Sales	Sales Department	2026-11-03 20:14:12.844471

	position_id	title	level	description
1	1	Intern	1	*NULL*
2	2	Junior Specialist	2	*NULL*
3	3	Specialist	3	*NULL*
4	4	Senior Specialist	4	*NULL*
5	5	Manager	5	*NULL*

	user_id	username	email	full_name
1	1	alice	alice@company.com	Alice Johnson
2	3	charlie	charlie@company.com	Charlie Brown
3	4	diana	diana@company.com	Diana Prince
4	5	eve	eve@company.com	Eve Wilson
5	2	bob	bob@company.com	Bob Smith

	project_id	name	description	owner_id	department_id	status	budget	start_date	end_date	created_at
1	1	Website Redesign	Redesign company website	1	1	active	50000.00	*NULL*	*NULL*	2026-11-03 20:14:12.853399
2	2	Mobile App	Develop mobile application	1	1	planning	100000.00	*NULL*	*NULL*	2026-11-03 20:14:12.853399
3	3	Test Project	Testing manager rights	1	1	active	*NULL*	*NULL*	*NULL*	2026-11-03 21:18:32.165177
4	4	Test Project	Testing manager rights	1	1	active	*NULL*	*NULL*	*NULL*	2026-11-03 21:19:18.733432

	task_id	project_id	title	description	status	priority	assignee_id	created_by	estimated_hours	actual_hours	due_date
1	2	1	Implement navigation	*NULL*	todo	3	3	1	*NULL*	*NULL*	*NULL*
2	3	1	Setup CI/CD	*NULL*	todo	2	3	1	*NULL*	*NULL*	*NULL*
3	4	1	Create color palette	*NULL*	review	3	2	1	*NULL*	*NULL*	*NULL*
4	5	1	Prepare responsive layout	*NULL*	todo	2	2	1	*NULL*	*NULL*	*NULL*
5	6	1	Test website	*NULL*	todo	2	3	1	*NULL*	*NULL*	*NULL*
6	7	2	Requirements analysis	*NULL*	in_progress	1	1	1	*NULL*	*NULL*	*NULL*
7	8	2	Architecture design	*NULL*	todo	1	3	1	*NULL*	*NULL*	*NULL*
8	9	2	Setup project structure	*NULL*	todo	2	2	1	*NULL*	*NULL*	*NULL*
9	10	2	Design login screen	*NULL*	todo	3	2	1	*NULL*	*NULL*	*NULL*

```
1  -- Тест 5: Создание комментария
2  INSERT INTO app.comments (task_id, user_id, content)
3  VALUES (1, 1, 'Test comment');
```

```
1
2  UPDATE app.tasks
3  SET status = 'in_progress'
4  WHERE task_id = 1;
```

Problems Output Debug Console Terminal Ports Postgre

Messages

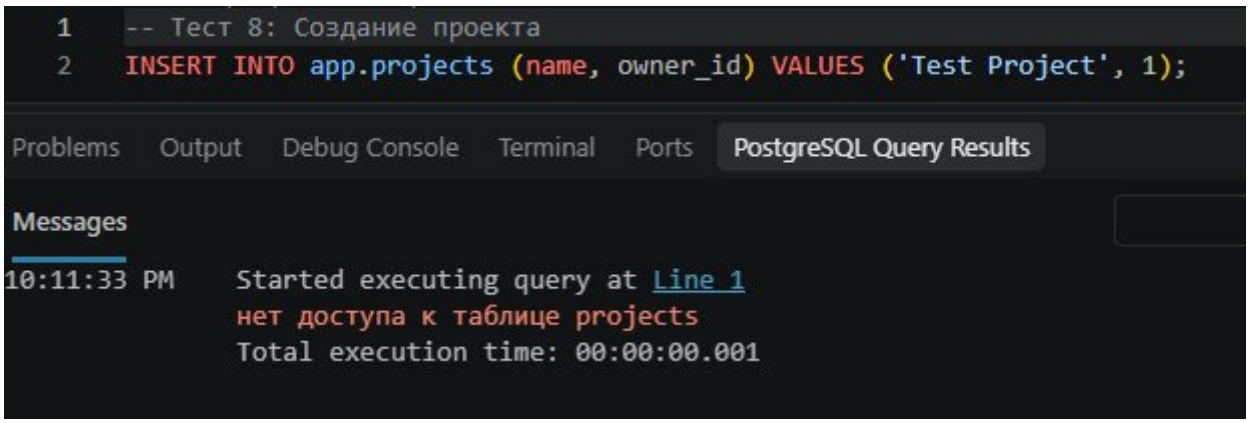
10:10:35 PM Started executing query at Line 2
UPDATE 1
Total execution time: 00:00:00.004

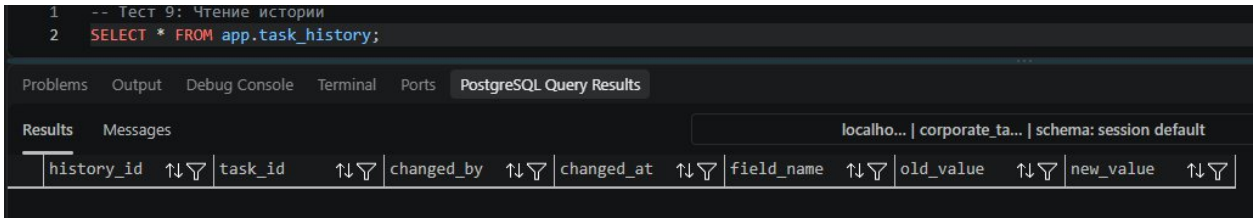
```
1  -- Тест 7: Удаление задачи
2  DELETE FROM app.tasks WHERE task_id = 1;
```

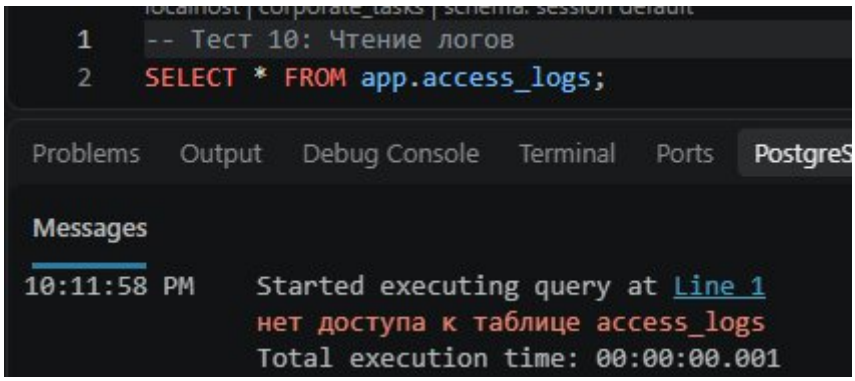
roblems Output Debug Console Terminal Ports Post

lessages

:11:12 PM Started executing query at Line 1
нет доступа к таблице tasks
Total execution time: 00:00:00


1 -- Тест 8: Создание проекта
2 INSERT INTO app.projects (name, owner_id) VALUES ('Test Project', 1);
Problems Output Debug Console Terminal Ports PostgreSQL Query Results
Messages
10:11:33 PM Started executing query at Line 1
нет доступа к таблице projects
Total execution time: 00:00:00.001


1 -- Тест 9: Чтение истории
2 SELECT * FROM app.task_history;
Problems Output Debug Console Terminal Ports PostgreSQL Query Results
Results Messages
localho... | corporate_ta... | schema: session default
history_id task_id changed_by changed_at field_name old_value new_value


1 -- Тест 10: Чтение логов
2 SELECT * FROM app.access_logs;
Problems Output Debug Console Terminal Ports
Messages
10:11:58 PM Started executing query at Line 1
нет доступа к таблице access_logs
Total execution time: 00:00:00.001

5.2

```
1  -- Тест 1: чтение справочников
2  SELECT * FROM app.departments;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho..

	department_id	name	description	created_at
1	1	IT Department	Information Technology	2026-11-03 20:14:12.844471
2	2	Marketing	Marketing and Communications	2026-11-03 20:14:12.844471
3	3	Sales	Sales Department	2026-11-03 20:14:12.844471

```
1  -- Тест 2: Создание проекта
2  INSERT INTO app.projects (name, description, owner_id, department_id, status)
3  VALUES ('Manager Test', 'Testing manager privileges', 1, 1, 'active');
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
10:17:21 PM    Started executing query at Line 1
               INSERT 0 1
               Total execution time: 00:00:00.006
```

localhost | corporate_tasks | schema: session default

```
1  -- Тест 3: Обновление проекта
2  UPDATE app.projects SET description = 'Updated description' WHERE name = 'Manager Test';
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages localho... |

```
10:17:37 PM    Started executing query at Line 1
               UPDATE 1
               Total execution time: 00:00:00.001
```

```
1  -- Тест 4: Удаление проекта
2  DELETE FROM app.projects
3  WHERE name = 'Manager Test';
```

roblems Output Debug Console Terminal Ports PostgreSC

lessages

```
:18:14 PM    Started executing query at Line 1
             нет доступа к таблице projects
             Total execution time: 00:00:00.001
```

```sql
-- Тест 5: Внутренние комментарии
SELECT comment_id, is_internal FROM app.comments;
```

Problems Output Debug Console Terminal Ports PostgreSQL Quer

Results Messages

	comment_id	is_internal
1	1	False
2	2	False
3	3	False
4	4	True
5	5	False
6	6	False

```sql
-- Тест 6: История изменений
SELECT * FROM app.task_history;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... | corporate_ta... | schema: session default

history_id	task_id	changed_by	changed_at	field_name	old_value	new_value

```sql
-- Тест 7: Чтение логов
SELECT * FROM app.access_logs;
```

Problems Output Debug Console Terminal Ports Postgre

Messages

```
10:22:09 PM    Started executing query at Line 1
               нет доступа к таблице access_logs
               Total execution time: 00:00:00.002
```

```sql
DELETE FROM app.projects
WHERE name = 'Manager Test';
```

Problems Output Debug Console Terminal Ports PostgreSQ

Messages

```
10:23:35 PM    Started executing query at Line 2
               DELETE 1
               Total execution time: 00:00:00.005
```

5.3

```sql
-- Тест 1: Чтение журнала
SELECT * FROM app.access_logs;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localho... | corporate_ta... | schema: session default

log_id	user_id	action	resource_...	resource_...	ip_address	logged_at

```sql
-- Тест 2: Полное управление задачами
DELETE FROM app.tasks WHERE task_id = 1;
INSERT INTO app.tasks (project_id, title, status, assignee_id, created_by)
VALUES (1, 'Admin Test Task', 'todo', 1, 1);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages localho... | corporate_ta... | schema: session default

```
Your transaction is currently aborted due to an error. You must explicitly issue a ROLLBACK or ROLLBACK TO SAVEPOINT before executing further commands.
Total execution time: 00:00:00
```

```sql
INSERT INTO app.tasks (
    project_id,
    title,
    status,
    assignee_id,
    created_by
)
VALUES (1, 'Admin Test Task', 'todo', 1, 1);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Resu

Messages

```
10:32:44 PM    Started executing query at Line 1
               INSERT 0 1
               Total execution time: 00:00:00.004
```

```sql
-- Тест 3: Создание новой таблицы
CREATE TABLE app.admin_test (id INT, name TEXT);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
10:29:48 PM    Started executing query at Line 1
               CREATE TABLE
               Total execution time: 00:00:00.008
```

```sql
-- Тест 4: удаление таблицы
DROP TABLE app.admin_test;
```

Problems Output Debug Console Terminal Ports PostgreSQ

Messages

```
0:33:20 PM    Started executing query at Line 1
              DROP TABLE
              Total execution time: 00:00:00.003
```

```sql
-- Тест 5: Проверка всех ролей
SELECT grantee, table_name, privilege_type
FROM information_schema.role_table_grants
WHERE grantee IN ('app_user', 'app_manager', 'app_admin')
ORDER BY grantee, table_name;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

grantee	table_name	privilege…

```sql
-- Тест 6: Изменение прав других ролей
GRANT SELECT ON TABLE app.access_logs TO app_manager;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages loca

```
0:33:57 PM    Started executing query at Line 1
              ПРЕДУПРЕЖДЕНИЕ: для объекта "access_logs" не были назначены никакие права
              GRANT
              Total execution time: 00:00:00
```

localhost | corporate_tasks | schema: session default

```sql
REVOKE SELECT ON TABLE app.access_logs FROM app_manager;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages localho... | corporate_ta... | schema: sess

```
10:34:42 PM    Started executing query at Line 2
               ПРЕДУПРЕЖДЕНИЕ: для объекта "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "tableoid" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "cmax" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "xmax" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "cmin" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "xmin" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "ctid" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "log_id" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "user_id" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "action" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "resource_type" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "resource_id" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "ip_address" отношения "access_logs" не были отозваны никакие права
               ПРЕДУПРЕЖДЕНИЕ: для столбца "logged_at" отношения "access_logs" не были отозваны никакие права
               REVOKE
               Total execution time: 00:00:00.001
```

6.1

```sql
-- 1. Матрица привилегий по ролям

SELECT
    grantee AS "Роль",
    table_name AS "Таблица",
    array_agg(privilege_type ORDER BY privilege_type) AS "Привилегии"
FROM information_schema.role_table_grants
WHERE grantee IN ('app_user', 'app_manager', 'app_admin')
GROUP BY grantee, table_name
ORDER BY grantee, table_name;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

Роль	Таблица	Привилегии

```sql
-- 2. Проверка иерархии

SELECT
    r.rolname AS "Роль",
    ARRAY_AGG(m.rolname) AS "Наследует от"
FROM pg_roles r
JOIN pg_auth_members am ON r.oid = am.member
JOIN pg_roles m ON am.roleid = m.oid
WHERE r.rolname IN ('app_user', 'app_manager', 'app_admin')
GROUP BY r.rolname
ORDER BY r.rolname;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	Роль	Наследует от
1	app_admin	{app_full_access,app_audit_read,app_manager}
2	app_manager	{app_project_write,app_history_read,app_user,app_i…
3	app_user	{app_comments_full,app_task_write,app_history_full…

```sql

-- 3. Default privileges

SELECT
    defaclrole::regrole AS "Для роли",
    defaclobjtype AS "Тип",
    defaclacl AS "Права"
FROM pg_default_acl
WHERE defaclnamespace::regnamespace::text = 'app';
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	Для роли	Тип	Права
1	app_admin	r	{app_read_reference=r/app_admin,app_task_write=arw…
2	app_admin	S	{app_task_write=U/app_admin,app_full_access=rwU/ap…

```sql

SET ROLE app_user;

SELECT
    has_table_privilege(
        current_user, 'app.users', 'SELECT'
    ) AS "Чтение пользователей",

    has_table_privilege(
        current_user, 'app.users', 'UPDATE'
    ) AS "Изменение пользователей",

    has_table_privilege(
        current_user, 'app.tasks', 'UPDATE'
    ) AS "Изменение задач",

    has_table_privilege(
        current_user, 'app.comments', 'DELETE'
    ) AS "Удаление комментариев";

RESET ROLE;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

localho... | corporate_ta... | schema: s

	Чтение пользователей	Изменение пользователей	Изменение задач	Удаление комментариев
1	True	False	True	True

```sql

SET ROLE app_manager;

SELECT
    has_table_privilege(
        current_user, 'app.projects', 'INSERT'
    ) AS "Создание проектов",

    has_table_privilege(
        current_user, 'app.access_logs', 'SELECT'
    ) AS "Чтение журнала";

RESET ROLE;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	Создание проектов	Чтение журнала
1	True	False

```sql

SET ROLE app_admin;

SELECT
    has_table_privilege(
        current_user, 'app.tasks', 'DELETE'
    ) AS "Удаление задач",

    has_schema_privilege(
        current_user, 'app', 'CREATE'
    ) AS "Создание объектов";

RESET ROLE;

```

Problems Output Debug Console Terminal Ports PostgreSQL

Results Messages

	Удаление задач	Создание объектов
1	True	True